

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Y Sravan Kumar(192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Y Sravan*

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

A company experiencing unpredictable traffic spikes during seasonal sales can use **cloud auto-scaling** and **load balancing** to maintain application performance. **Auto-scaling** automatically increases the number of server instances when traffic rises and removes extra instances when demand decreases. **Load balancing** distributes incoming user requests evenly across all available servers, preventing any single server from becoming overloaded.

This approach **reduces downtime** by ensuring that enough resources are always available to handle user requests. It also **improves the user experience** by providing faster response

times, reducing page loading delays, and allowing customers to complete purchases without interruptions, even during peak traffic.

Monitoring tools play an important role by continuously tracking CPU usage, memory, network traffic, and application performance. When resource usage reaches a predefined limit, they trigger auto-scaling and send alerts if any server or application fails. This helps administrators detect and resolve issues before they affect users.

Real-world example: During **Amazon Prime Day** or **Flipkart Big Billion Days**, millions of users visit the website at the same time. Auto-scaling launches additional servers, load balancing spreads traffic evenly across them, and monitoring tools ensure the platform remains stable and responsive throughout the event.

Conclusion: Using **auto-scaling, load balancing, and monitoring tools** enables e-commerce companies to handle sudden traffic spikes efficiently, minimize downtime, and deliver a smooth shopping experience.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Cloud Storage for Backup Data

- **Durability and Data Redundancy**
 - Data is stored in multiple copies across different servers or regions.
 - If one server fails, another copy is available, ensuring high durability.
- **Encryption and Access Control**
 - Encrypt data at rest and in transit.
 - Use **IAM, RBAC, and MFA** so only authorized users can access backup data.
- **Cost Advantages**
 - Pay only for the storage used.
 - No need to buy or maintain physical storage hardware.
 - Move old backups to low-cost archival storage.
- **Disaster Recovery Example**

- If a company's data center is damaged by a fire or flood, data can be restored from cloud backups, allowing business operations to continue quickly.

Conclusion: Cloud storage provides **secure, reliable, and cost-effective** backup with high durability, data redundancy, and fast disaster recovery.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Global Deployment Using CDN

- **Content Delivery Network (CDN)**

- Stores app content on servers located around the world.
- Delivers content from the nearest server, reducing latency and improving loading speed.

- **Role of Edge Servers**

- Edge servers cache images, videos, and other static content close to users.
- This reduces buffering, speeds up downloads, and provides a smoother user experience across different regions.

- **Global Availability**

- Cloud providers use multiple **regions** and **availability zones**.
- If one region fails, traffic is redirected to another region, ensuring continuous service and high availability.

- **Real-World Example**

- **Netflix** uses CDNs and edge servers to stream videos with minimal buffering worldwide.
- Similarly, online games such as **PUBG** or **Fortnite** use edge servers to reduce latency and provide smooth gameplay.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data

sovereignty requirements are met? Give an example involving sensitive financial transactions.

Cloud Security and Compliance for Financial Institutions

- **Regulatory Compliance and Auditing**
 - Cloud providers support compliance through security standards, audit reports, and regular compliance checks.
 - Audit logs help track all user activities and financial transactions.
- **Logging, Monitoring, and Encryption**
 - **Logging** records user actions and system events.
 - **Monitoring** detects unusual activities and security threats in real time.
 - **Encryption** protects data both **at rest** and **in transit**, preventing unauthorized access.
- **Data Sovereignty**
 - Store financial data in cloud regions located within the required country or region.
 - Use access policies to ensure data does not leave approved locations.
- **Example**
 - A bank stores customer transaction records in a cloud data center within its country. All transactions are encrypted, user access is controlled through **IAM** and **MFA**, and audit logs record every transaction to meet regulatory requirements.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Containerization for Application Deployment

- **Containers vs Virtual Machines**
 - **Containers** share the host operating system, making them lightweight, faster, and use fewer resources.
 - **Virtual Machines (VMs)** have their own operating system, so they consume more CPU, memory, and storage.

- **Container Orchestration (Kubernetes)**
 - Kubernetes automates container deployment and management.
 - It supports **auto-scaling**, load balancing, self-healing, and automatic updates, making applications easier to manage.
- **Portability**
 - Containers package the application with its dependencies, allowing it to run consistently across different cloud providers and operating systems without modification.
- **Example**
 - An **e-commerce application** built with microservices, where separate containers handle **user login, product catalog, shopping cart, payment, and order management**. Each service can be deployed and scaled independently.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Assessment Tasks – Application Based Questions

1. Seasonal Sales – Auto-Scaling and Load Balancing

Auto-Scaling

- Automatically adds server instances when traffic increases.
- Removes unnecessary instances when traffic decreases.
- Ensures sufficient resources during peak demand.
- Helps reduce costs during low-traffic periods.

Load Balancing

- Distributes incoming requests across multiple servers.
- Prevents a single server from becoming overloaded.
- Routes traffic away from unhealthy servers.

Reducing Downtime and Improving User Experience

- Extra resources prevent application crashes during traffic spikes.
- Users experience faster page loading and response times.
- Customers can browse products and complete payments without interruptions.

Role of Monitoring

- Continuously monitors CPU, memory, network traffic, response time, and errors.
- Can trigger auto-scaling when predefined thresholds are reached.
- Generates alerts when failures or unusual behavior are detected.

Real-World Example

During Amazon Prime Day or Flipkart Big Billion Days, millions of customers may access the platform simultaneously. Auto-scaling increases resources, load balancing distributes requests, and monitoring helps maintain application performance.

Conclusion: Auto-scaling, load balancing, and monitoring help e-commerce platforms handle sudden traffic efficiently while maintaining availability and performance.

2. Cloud Storage for Backup Data

Durability and Data Redundancy

- Cloud providers maintain multiple copies of data.
- Copies may be distributed across different servers or availability zones.
- If one storage device fails, another copy remains available.
- This provides high durability and reliability.

Encryption and Access Control

- Encryption at rest protects stored backup data.
- Encryption in transit protects data during transfer.
- IAM controls who can access the backups.
- RBAC provides permissions based on user roles.
- MFA provides an additional layer of authentication.

Cost Advantages

- Organizations pay for the storage they use.
- No need to purchase and maintain large physical storage systems.
- Old backups can be moved to low-cost archive storage.
- Storage capacity can be increased without purchasing new hardware.

Disaster Recovery Example

If an organization's on-premises data center is damaged by a fire, flood, or hardware failure, applications and databases can be restored using cloud backups.

Conclusion: Cloud storage provides durable, secure, scalable, and cost-effective backup and disaster recovery.

3. Global Mobile Application – CDN

Content Delivery Network (CDN)

- A CDN stores cached content at geographically distributed locations.
- Content is delivered from a location closer to the user.
- This reduces network latency.
- It improves application and website loading speed.

Edge Servers

- Edge servers operate closer to end users.
- They cache images, videos, downloads, and other suitable content.
- Reduce the distance data must travel.
- Reduce buffering and improve responsiveness.

Global Availability

- Cloud providers operate infrastructure across multiple geographic regions.
- Applications can be deployed closer to users.
- Multiple availability zones improve resilience within a region.
- Multi-region architectures can provide failover if an entire region becomes unavailable.

Real-Time Example

A global video-streaming service can use CDN edge locations to deliver video content closer to viewers, reducing buffering. Online gaming platforms can similarly use geographically distributed infrastructure to reduce latency.

Conclusion: CDNs, edge infrastructure, and multi-region cloud deployment provide faster and more reliable services to global users.

4. Financial Institution – Security and Compliance

Regulatory Compliance and Auditing

- Cloud providers provide security controls, compliance documentation, and audit reports.
- Organizations must configure their cloud environments according to applicable regulations.
- Audit trails record important user and administrative activities.
- Regular audits help identify compliance violations.

Logging

- Records login attempts, system events, data access, and administrative actions.
- Helps investigate security incidents.
- Provides evidence for audits.

Monitoring

- Continuously monitors systems for unusual activities.
- Detects potential security threats and failures.
- Alerts administrators when suspicious behavior occurs.

Encryption

- Data at rest should be encrypted in databases, storage, and backups.
- Data in transit should be protected using secure protocols such as TLS.
- Encryption keys should be securely managed.

Data Sovereignty

- Store regulated data only in approved geographic regions.
- Apply policies controlling where data and backups are stored.
- Ensure replication does not unintentionally move regulated data to prohibited locations.

Example

A bank can store customer transaction information in an approved cloud region. Transactions are encrypted, IAM controls access, MFA protects sensitive accounts, and audit logs record access and administrative operations.

Conclusion: Strong IAM, encryption, monitoring, auditing, and geographic controls help financial institutions meet security and compliance requirements.

5. Containerization for Application Deployment

Containers vs Virtual Machines

Containers:

- Share the host operating-system kernel.
- Lightweight and require fewer resources.
- Start quickly.
- Allow many containers to run on the same infrastructure.

Virtual Machines:

- Each VM generally includes its own guest operating system.
- Require more CPU, memory, and storage.
- Usually take longer to start.
- Provide virtualization at the hardware level.

Kubernetes Orchestration

Kubernetes simplifies container management by providing:

- Automated deployment
- Auto-scaling
- Load balancing
- Self-healing

- Rolling updates
- Service discovery

If a container fails, Kubernetes can automatically replace it.

Portability

- Containers package an application together with its required dependencies.
- The same container image can run consistently across compatible environments.
- This makes it easier to move applications between on-premises systems and different cloud providers.

Microservices Example

An e-commerce application can contain separate services for:

- User Login
- Product Catalog
- Shopping Cart
- Payment
- Order Management

Each microservice can run in a separate container and can be deployed, updated, and scaled independently.

Conclusion: Containers provide lightweight and portable application deployment, while Kubernetes simplifies their deployment, scaling, recovery, and management.